«part def»

Specified Nanosatellite EPS Engineering Framework :> 'Measured Nanosatellite EPS Engineering Framework'

doc

Complete OOSEM black-box definition. Inherited features provide six boundary service functions, six typed ports, eleven measures of performance, and the operational control state machine. The features below add required stores and externally declared execution resource characteristics. Internal components remain unspecified.

attributes

^<MOP-01> valid trace coverage : Percent
^<MOP-02> reference analysis reproduction success : Percent
^<MOP-03> open critical discrepancy count : Count
^<MOP-04> completed contrasting case count : Count
^<MOP-05> undeclared adaptation count : Count
^<MOP-06> correct change turnaround time : Hours
^<MOP-07> case adaptation effort : PersonHours
^<MOP-08> independent reproduction effort : PersonHours
^<MOP-09> framework adoption effort : PersonHours
^<MOP-10> direct framework cost : CAD
^<MOP-11> consistency assurance effectiveness : Percent
declared minimum execution environment : String
declared runtime dependencies : String

«ref part»

^systemModel : 'Versioned System Model'

ports

^frameworkControl : FrameworkBoundaryInterfaces::'Model Control Port'

«ref part»

^evidenceRepository : 'Evidence Repository'

ports

^frameworkCustody : FrameworkBoundaryInterfaces::'Evidence Custody Port'

«ref part»

^analysisTools : 'Analysis Tools'

ports

^frameworkExecution : FrameworkBoundaryInterfaces::'Analysis Execution Port'

«perform action»

^maintainEvidenceChain : FrameworkBlackBoxFunctions::'Maintain Evidence Chain'

items

^request : 'Evidence Chain Request'
^modelRecords : 'Controlled Model Records'
^evidenceRecords : 'Retained Evidence Records'
^scope : 'Evidence Chain Scope'
^assessment : 'Evidence Chain Assessment'

actions

^scopeEvidenceChain : 'Scope Evidence Chain'
^validateEvidenceChain : 'Validate Evidence Chain'
^issueAssessment : 'Issue Evidence Chain Assessment'

«#requiredStore ref item»

requiredRecords : 'Controlled Framework Record Set'

items

^modelRecords : 'Controlled Model Records' [0..*]
^evidenceRecords : 'Retained Evidence Records' [0..*]
^caseConfigurations : 'Controlled Case Configuration' [0..*]
^evaluationManifest : 'Framework Evaluation Manifest'
^chainAssessments : 'Evidence Chain Assessment' [0..*]
^reproductionVerdicts : 'Reproduction Verdict' [0..*]
^changePackages : 'Change Evidence Package' [0..*]
^comparisonPackages : 'Candidate Comparison Package' [0..*]
^reviewPackages : 'Review Package' [0..*]
^reviewStatuses : 'Review Status' [0..*]
^transferVerifications : FrameworkBoundaryInterfaces::'Transfer Verification Record' [0..*]

«perform action»

^configureFrameworkForCase : FrameworkBlackBoxFunctions::'Configure Framework for EPS Case'

items

^request : 'Case Configuration Request'
^commonCore : 'Common Core Package'
^adaptationDecision : 'Adaptation Decision'
^draft : 'Case Configuration Draft'
^configuration : 'Controlled Case Configuration'
^status : 'Configuration Status'

actions

^captureConfiguration : 'Capture Case Configuration'
^validateConfiguration : 'Validate Case Configuration'
^issueStatus : 'Issue Configuration Status'

«perform action»

^reproduceDesignatedAnalysis : FrameworkBlackBoxFunctions::'Reproduce Designated Analysis'

items

^request : 'Reproduction Request'
^modelRecords : 'Controlled Model Records'
^evidenceRecords : 'Retained Evidence Records'
^returnedResult : 'Analysis Result Package'
^executionRequest : 'Analysis Execution Request'
^result : 'Analysis Result Package'
^verdict : 'Reproduction Verdict'

actions

^preparePackage : 'Prepare Reproduction Package'
^coordinateAnalysis : 'Coordinate Controlled Analysis'
^evaluateResult : 'Evaluate Reproduced Result'
^issueVerdict : 'Issue Reproduction Verdict'

«perform action»

^supportEarlyDesignReview : FrameworkBlackBoxFunctions::'Support Early Design Review'

items

^request : 'Design Review Request'
^modelRecords : 'Controlled Model Records'
^evidenceRecords : 'Retained Evidence Records'
^findings : 'Review Findings'
^disposition : 'Finding Disposition'
^outcome : 'Review Outcome'
^reviewPackage : 'Review Package'
^issuedFindings : 'Review Findings'
^reproduction : 'Reproduction Verdict'
^status : 'Review Status'

actions

^assemblePackage : 'Assemble Review Package'
^coordinateReproduction : 'Coordinate Selected Reproduction'
^manageFindings : 'Manage Review Findings'
^issueStatus : 'Issue Review Status'

«perform action»

^assessControlledDesignChange : FrameworkBlackBoxFunctions::'Assess Controlled Design Change'

items

^request : 'Design Change Request'
^modelRecords : 'Controlled Model Records'
^evidenceRecords : 'Retained Evidence Records'
^revisions : 'Change Revision Package'
^returnedResult : 'Analysis Result Package'
^executionRequest : 'Analysis Execution Request'
^impact : 'Change Impact Assessment'
^evidencePackage : 'Change Evidence Package'

actions

^validateRequest : 'Validate Change Request'
^analyzeImpact : 'Analyze Change Impact'
^coordinateAnalysis : 'Coordinate Controlled Analysis'
^validateEvidence : 'Validate Regenerated Evidence'
^assembleEvidence : 'Assemble Change Evidence Package'

«perform action»

^compareCandidateArchitectures : FrameworkBlackBoxFunctions::'Compare Candidate EPS Architectures'

items

^request : 'Candidate Comparison Request'
^protocol : 'Comparison Protocol'
^candidates : 'Candidate Model Package'
^returnedResults : 'Analysis Result Package'
^sensitivityAndUncertainty : 'Sensitivity and Uncertainty Assessment'
^executionRequest : 'Analysis Execution Request'
^result : 'Analysis Result Package'
^comparisonPackage : 'Candidate Comparison Package'

actions

^validateBasis : 'Validate Comparison Basis'
^coordinateAnalysis : 'Coordinate Controlled Analysis'
^determineFeasibility : 'Determine Candidate Feasibility'
^compareCandidates : 'Compare Candidate Architectures'
^issueComparison : 'Issue Candidate Comparison'

«exhibit state»

^operationalControl : FrameworkOperationalState::'Framework Operational Control'

states

^<ST-01> Unconfigured
^<ST-02> Configuring
^<ST-03> Controlled Baseline Ready
^<ST-04> Assessing Evidence
^<ST-05> Reproducing Analysis
^<ST-06> Assessing Change
^<ST-07> Comparing Candidates
^<ST-08> Under Review
^<ST-09> Discrepancy Open
^<ST-10> Recovering

state transition

^initializeControl
^<T-01> beginConfiguration
^<T-02> configurationAccepted
^<T-03> configurationDeficient
^<T-04> reconfigureContext
^<T-05> assessEvidenceRequest
^<T-06> reproduceAnalysisRequest
^<T-07> assessChangeRequest
^<T-08> compareCandidatesRequest
^<T-09> reviewRequest
^<T-10> evidenceAccepted
^<T-11> evidenceDeficient
^<T-12> reproductionAccepted
^<T-13> reproductionDeficient
^<T-14> changeAccepted
^<T-15> changeDeficient
^<T-16> comparisonAccepted
^<T-17> comparisonDeficient
^<T-18> reviewAccepted
^<T-19> reviewDeficient
^<T-20> beginControlledRecovery
^<T-21> retryConfiguration
^<T-22> retryReproduction
^<T-23> retryChange
^<T-24> retryComparison
^<T-25> retryReview
^<T-26> recoveryAccepted
^<T-27> recoveryDeficient

^missionApplication : FrameworkBoundaryInterfaces::'Mission Application Port'
^researchGovernance : FrameworkBoundaryInterfaces::'Research Governance Port'
^independentReview : FrameworkBoundaryInterfaces::'Independent Review Port'
^modelControl : FrameworkBoundaryInterfaces::'Model Control Port'
^analysisExecution : FrameworkBoundaryInterfaces::'Analysis Execution Port'
^evidenceCustody : FrameworkBoundaryInterfaces::'Evidence Custody Port'